

AquaTrace

AI-Assisted Water Leak Detection, Localization and Inspection Prioritization

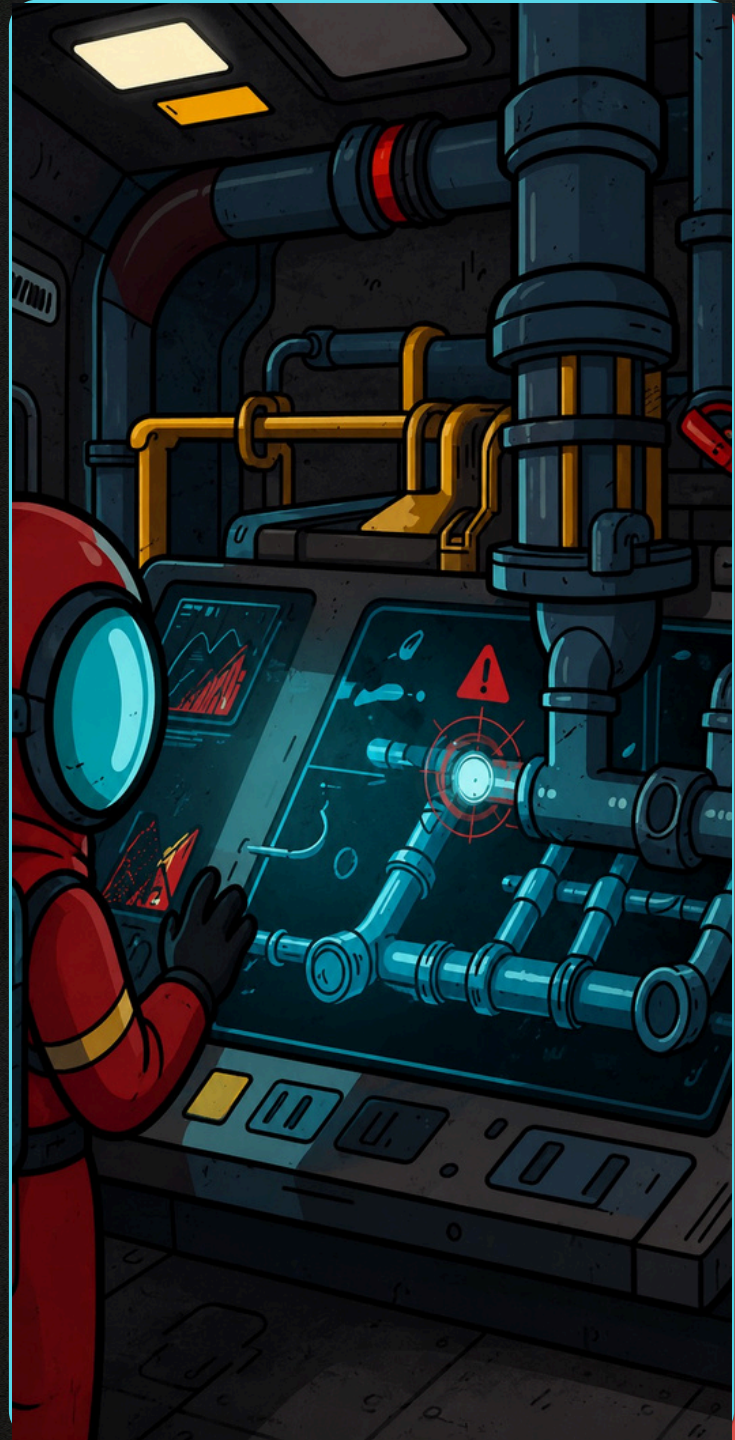
HACKATHON TRACK: TECH INNOVATION

TEAM NAME: **PreciseX**

Problem Statement – The Hidden Impostor

Problem: Water leaks can remain undetected because abnormal flow and pressure readings do not always reveal the exact location or cause of the problem.

- Hidden Water Loss: Small leaks may continue for hours before being noticed.
- Difficult Localization: Identifying the affected pipeline section can require time-consuming inspections.
- Misleading Readings: Faulty sensors and sudden changes in water demand can resemble leaks.
- Limited Resources: Small buildings and facilities may not be able to afford extensive monitoring equipment.



Why Is This Important?

Undetected leaks waste water, increase maintenance costs and can damage infrastructure. Unnecessary inspections also consume time and resources.

Proposed Solution – AquaTrace

AquaTrace is an AI-assisted dashboard that identifies suspicious pipeline sections ,compares possible causes and helps users decide what to inspect first.

How It Works:

01

Monitor:

Collect flow and pressure readings from sensors or simulated data.

02

Detect:

Identify unusual readings by comparing them with expected patterns.

03

Investigate:

Rank suspected pipeline sections using available readings and pipeline connections.

04

Prioritize:

Recommend which section to inspect first based on suspected severity, possible water loss and inspection effort.

Expected Outcome: Faster investigation, fewer unnecessary inspections and improved visibility into potential water loss.



How AquaTrace Works

01

Step 1: Input Data

- Flow rate
- Water pressure
- Pipeline connections

02

Step 2: Detect Anomalies

- Compare readings with expected values.
- Identify unusual changes in flow or pressure.

03

Step 3: Rank Pipeline Sections

- Compare sensor readings and connected pipes.
- Identify and rank the most likely leak locations.

04

Step 4: Recommend Action

- Display thesuspected location.
- Estimate possible water loss.
- Suggest which section should be inspected first.

Illustrative Dashboard Output

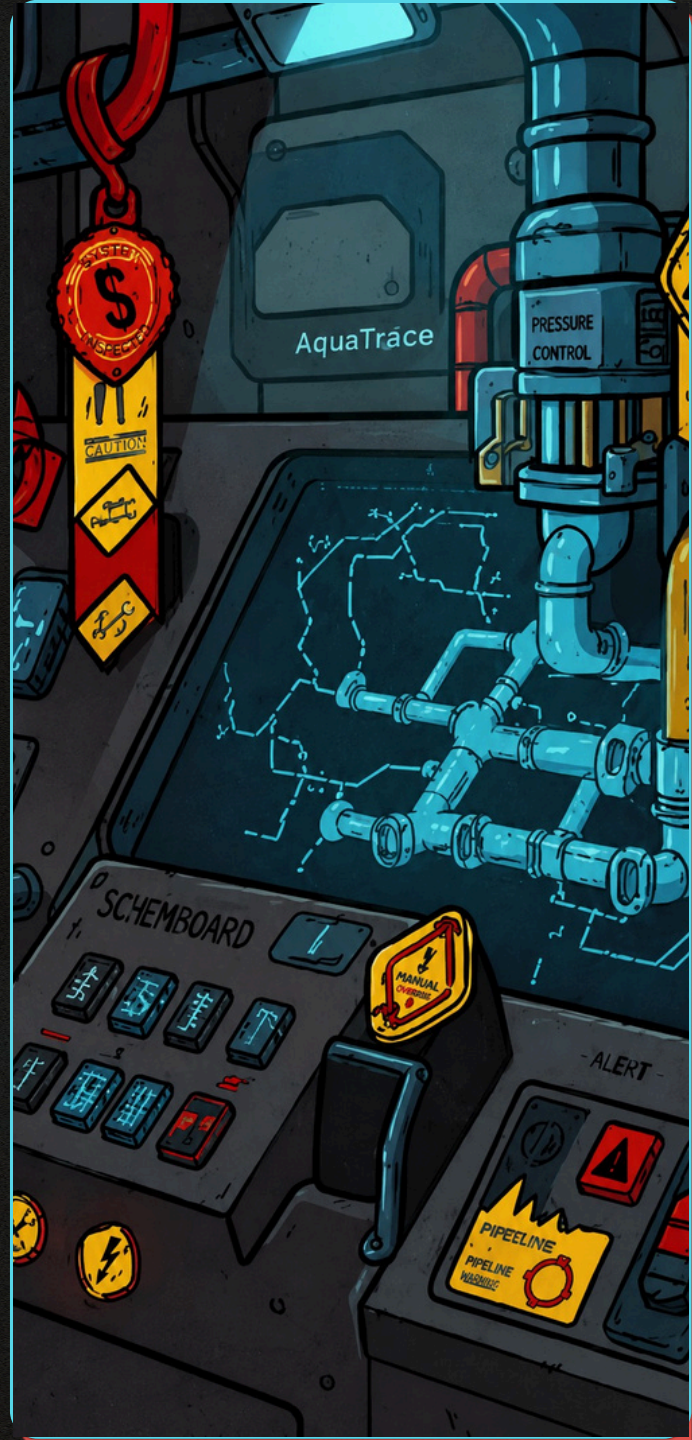
Pipeline Section	Status	Suggested Action
Section A	Normal	Continue monitoring
Section B	Possible leak	Inspect first
Section C	Sensor reading uncertain	Verify sensor
Section D	Normal	Continue monitoring



Unique Features and Differentiation

Our goal is not just to detect a leak, but to help users investigate the right problem first.

1. **Low-Sensor Investigation** — Test whether suspected leaks can still be ranked when only a limited number of sensors are available.
2. **Leak vs Sensor Fault vs Demand Change** — Compare three scenarios: a real leak, a faulty sensor and a sudden increase in normal water use. This helps prevent every abnormal reading from automatically being labelled a leak.
3. **Smart Inspection Priority** — Rank suspected sections using leak likelihood, estimated severity and inspection effort.
4. **Water-Loss Estimator** — Estimate possible unaccounted water volume over time and show the calculation transparently.
5. **Interactive Leak Simulator** — Allow users to trigger different scenarios and watch the pipeline map, alerts and rankings change without needing physical sensors.

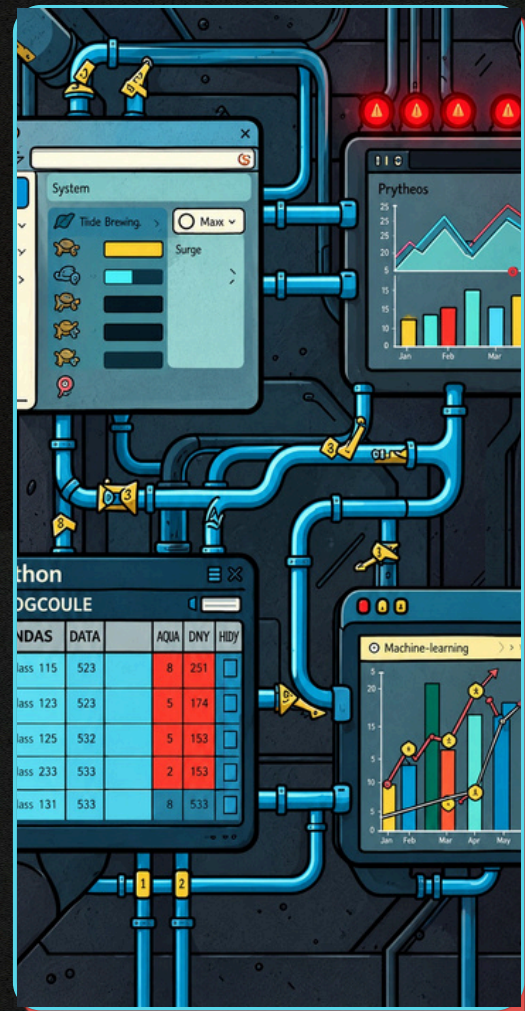


Technical Approach and Technology Stack

1. **Frontend** — HTML, CSS and JavaScript — Build the dashboard, pipeline diagram, buttons, alerts and charts.
2. **Backend** — Python — Process readings, check for unusual values, calculate possible water loss and rank suspected sections.
3. **Data Handling** — Pandas — Organize readings into tables and compare measurements.
4. **Charts** — Chart.js — Visualize flow, pressure, estimated water loss and alert history.
5. **Optional Machine Learning** — Scikit-learn — Train a basic model on simulated scenarios after the rule-based version works.

Implementation Plan

- Generate simulated flow and pressure readings.
- Start with simple rules and thresholds for anomaly detection.
- Add pipeline-section ranking and water-loss calculations.
- Create an interactive dashboard with scenario controls.
- Add a basic machine-learning model if time permits.



Technical Limitation: The first version is a simulation-based proof of concept. Real-world deployment would require actual sensor data and field testing.

Feasibility, Validation and Potential Impact

Feasibility: AquaTrace can be developed as a small software prototype using simulated data, basic detection rules and an interactive dashboard.

How Will We Test It?

- Detection Performance: Measure how many simulated leak scenarios are identified correctly.
- Localization Accuracy: Check how often the actual affected section ranks first.
- False Alarms: Measure how often sensor faults or demand changes are incorrectly flagged as leaks.
- Robustness: Test the effect of noisy readings and missing sensors.

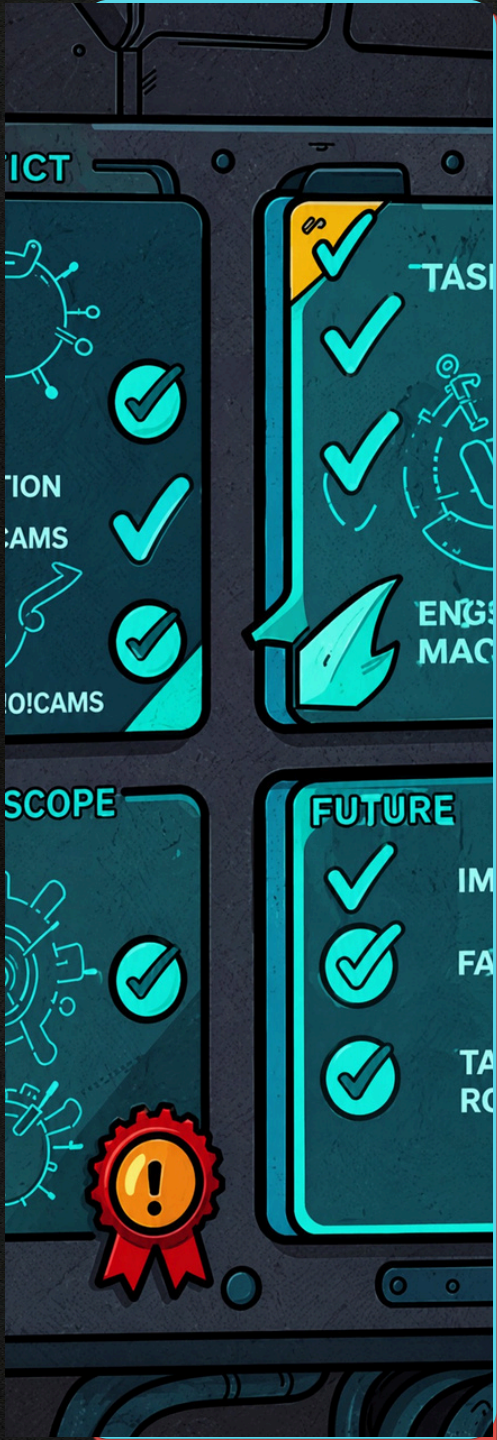
Potential Impact

- Reduce time spent investigating suspected leaks.
- Help campuses, buildings and small facilities prioritize maintenance.
- Improve awareness of potential water loss.
- Provide a low-cost starting point before investing in physical monitoring equipment.

Future Scope

- Connect real flow and pressure sensors.
- Use more advanced machine-learning models.
- Add notifications and maintenance reports.
- Test the approach on larger pipeline networks.

Expected Impact: Better-informed inspections and more efficient water-loss investigation. Actual savings and performance will need to be measured through testing.




THANK YOU!
AquaTrace
Find the Leak. Save Every Drop.